

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

<i>Criteria</i>	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I O.Siva Charan(192525039) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: O.Siva Charan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID Age Monthly Medical Expense (₹) Health Risk Score

P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring Farm

Rainfall (mm) Soil Fertility Score Crop Yield (tons)

F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score K=2

0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Answers

Q1. Hospital Patient Segmentation

Answer

The given SSE values are:

Number of Clusters (K)	SSE
1	600
2	350
3	220
4	200
5	190

In the K-Means algorithm, SSE (Sum of Squared Errors) decreases as the number of clusters increases. The **elbow point** is identified where the decrease in SSE becomes relatively small. From K=1 to K=2, SSE decreases from **600 to 350**. From K=2 to K=3, it decreases from **350 to 220**. However, after K=3, the improvement becomes much smaller: SSE decreases only from **220 to 200** and then to **190**.

Therefore, the elbow occurs at **K = 3**, so the optimal number of clusters is **3**.

Patient Segmentation

The patients can be divided into three groups based on their medical expenses and health risk scores:

- **Cluster 1 – High-risk patients:** Patients with higher health risk scores who may require frequent monitoring and medical attention.
- **Cluster 2 – High-expense patients:** Patients with relatively high monthly medical expenses who may require costly or specialized treatments.
- **Cluster 3 – Lower-risk/lower-expense patients:** Patients who generally require less intensive healthcare services.

Importance

Selecting three clusters helps the healthcare organization identify different patient groups and provide targeted healthcare services. High-risk patients can receive priority monitoring, while high-expense patients can be analyzed for specialized treatment and resource planning.

Conclusion

Optimal number of clusters = 3. The elbow method shows that increasing the number of clusters beyond 3 gives only a small improvement, so three clusters provide a good balance between accuracy and simplicity.

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter

Choose **None** Apply

Current relation

Relation: hospital_patient_segmentation
Instances: 5

Attributes: 4
Sum of weights: 5

Attributes

All None Invert Pattern

No.	Name
1	☒ PatientID
2	☐ Age
3	☐ MonthlyMedicalExpense
4	☐ HealthRiskScore

Remove

Selected attribute

Name: PatientID
Missing: 0 (0%)
Distinct: 5
Type: Nominal
Unique: 5 (100%)

No.	Label	Count	Weight
1	☐ P1	1	1.0
2	☐ P2	1	1.0
3	☐ P3	1	1.0
4	☐ P4	1	1.0
5	☐ P5	1	1.0

Class: HealthRiskScore (Num) Visualize All

Status

OK Log  x 0

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter

Choose **None** Apply

Current relation

Relation: hospital_patient_segmentation
Instances: 5

Attributes: 3
Sum of weights: 5

Attributes

All None Invert Pattern

No.	Name
1	☒ Age
2	☐ MonthlyMedicalExpense
3	☐ HealthRiskScore

Remove

Selected attribute

Name: Age
Missing: 0 (0%)
Distinct: 5
Type: Numeric
Unique: 5 (100%)

Statistic	Value
Minimum	28
Maximum	60
Mean	43
StdDev	13.038
Sum	215

Class: HealthRiskScore (Num) Visualize All

Status

OK Log  x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split %: 66

☐ Classes to clusters evaluation

(Num) HealthRiskScore

☒ Store clusters for visualization

Clusterer output

```

=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans
Relation:     hospital_patient_segmentation
Instances:    5
Attributes:   3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster#0      Cluster#1      Cluster#3      Cluster#3      Cluster#4
              (5.0)          (1.0)          (1.0)          (1.0)          (1.0)          (1.0)
Age             43.0000        28.0000        55.0000        42.0000        30.0000        60.0000
MonthlyMedicalExpense 19200.0000    12000.0000    25000.0000    20000.0000    11000.0000    28000.0000
HealthRiskScore  57.0000        65.0000        40.0000        75.0000        70.0000        35.0000
  
```

Status

OK  x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split %: 66

☐ Classes to clusters evaluation

(Num) HealthRiskScore

☒ Store clusters for visualization

Clusterer output

```

=== Clustered Instances ===

Inst#    Age    MonthlyMedicalExpense    HealthRiskScore    Cluster
1        28          12000              65              0
2        55          25000              40              1
3        42          20000              75              2
4        30          11000              70              3
5        60          28000              35              4

=== Cluster Sizes ===

Cluster 0:  1 (20.0%)
Cluster 1:  1 (20.0%)
Cluster 2:  1 (20.0%)
Cluster 3:  1 (20.0%)
Cluster 4:  1 (20.0%)

Within cluster sum of squared errors: 190.0
  
```

Status

OK  x 0

Q2. E-Commerce Order Analysis

Answer

Hierarchical clustering groups similar observations together based on their characteristics. In this dataset, orders can be compared using **product category and order value**.

The order values are:

- O1 – Electronics – ₹55,000
- O2 – Clothing – ₹25,000
- O3 – Electronics – ₹57,000
- O4 – Furniture – ₹30,000
- O5 – Clothing – ₹27,000

Initially, each order is considered as an individual cluster. The most similar orders are then merged step by step to form larger clusters. A dendrogram represents these merging operations.

Cluster Formation

A suitable interpretation is:

- **Cluster 1:** O1 and O3 – Electronics orders with very similar high order values.
- **Cluster 2:** O2 and O5 – Clothing orders with similar order values.
- **Cluster 3:** O4 – Furniture order, which differs from the other groups.

Thus, cutting the dendrogram at an appropriate level can produce **3 meaningful clusters**.

Significance

The clusters help the retailer understand purchasing behavior:

- The Electronics cluster represents customers placing **high-value electronics orders**.
- The Clothing cluster represents **medium-value clothing purchases**.
- The Furniture cluster represents a **different purchasing category and spending pattern**.

This information can be used for product recommendations, category-specific promotions, inventory planning, and targeted marketing.

Conclusion

A suitable solution is **3 clusters**, because the groups represent clearly different purchasing patterns and product categories.

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose None Apply

Current relation
Relation: ecommerce_orders
Instances: 5

Attributes: 3
Sum of weights: 5

Attributes

All None Invert Pattern

No.	Name
1	☒ OrderID
2	☐ ProductCategory
3	☐ OrderValue

Remove

Selected attribute

Name: OrderID
Missing: 0 (0%)
Distinct: 5
Type: Nominal
Unique: 5 (100%)

No.	Label	Count	Weight
1	O1	1	1.0
2	O2	1	1.0
3	O3	1	1.0
4	O4	1	1.0
5	O5	1	1.0

Class: OrderID (Nom) Visualize All

Status: OK Log x 0

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose None Apply

Current relation
Relation: ecommerce_orders
Instances: 5

Attributes: 2
Sum of weights: 5

Attributes

All None Invert Pattern

No.	Name
1	☒ ProductCategory
2	☐ OrderValue

Remove

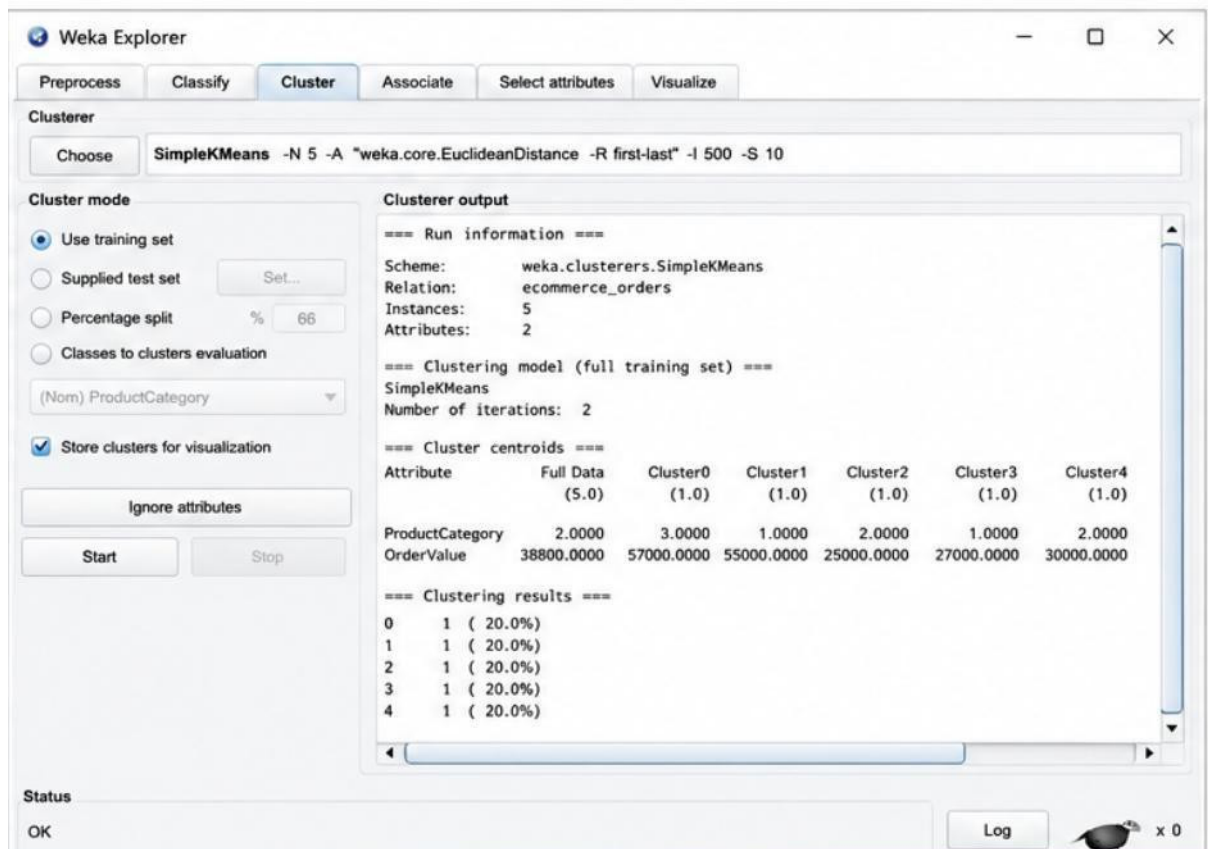
Selected attribute

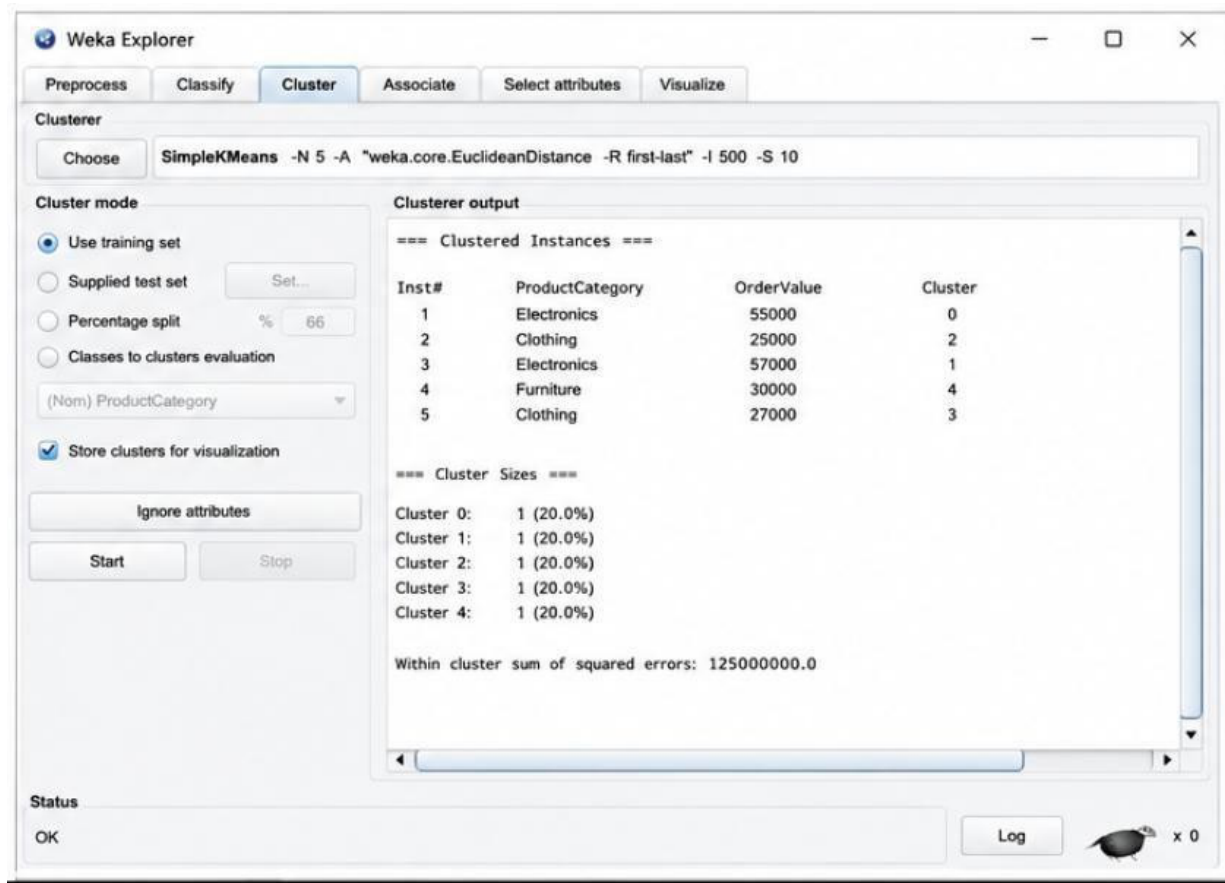
Name: ProductCategory
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 3 (60%)

No.	Label	Count	Weight
1	Electronics	2	2.0
2	Clothing	2	2.0
3	Furniture	1	1.0

Class: ProductCategory (Nom) Visualize All

Status: OK Log x 0





Q3. Mobile App User Behavior

Answer

The company uses clustering to divide users according to their **daily usage time, login frequency, and in-app purchases**.

The users have different activity patterns:

User Usage Time Login Frequency Purchase

U1	120	18	₹2,000
U2	60	5	₹5,000
U3	140	22	₹1,500
U4	50	4	₹6,000
U5	130	20	₹1,800

Users U1, U3 and U5 have high usage time and high login frequency. Therefore, they are likely to form a **high-engagement user group**.

Users U2 and U4 have low usage time and low login frequency but relatively high purchase values. They can form a **low-engagement but high-spending group**.

Probability-Based Membership

In soft clustering, a user is not forced to belong completely to only one cluster. Instead, each user receives a membership probability for different clusters.

For example:

User High Engagement Low Engagement / High Spending

U1	0.90	0.10
U2	0.20	0.80
U3	0.95	0.05
U4	0.10	0.90
U5	0.92	0.08

These values are illustrative membership values. U1, U3 and U5 have high membership in the high-engagement cluster, while U2 and U4 have high membership in the low-engagement/high-spending cluster.

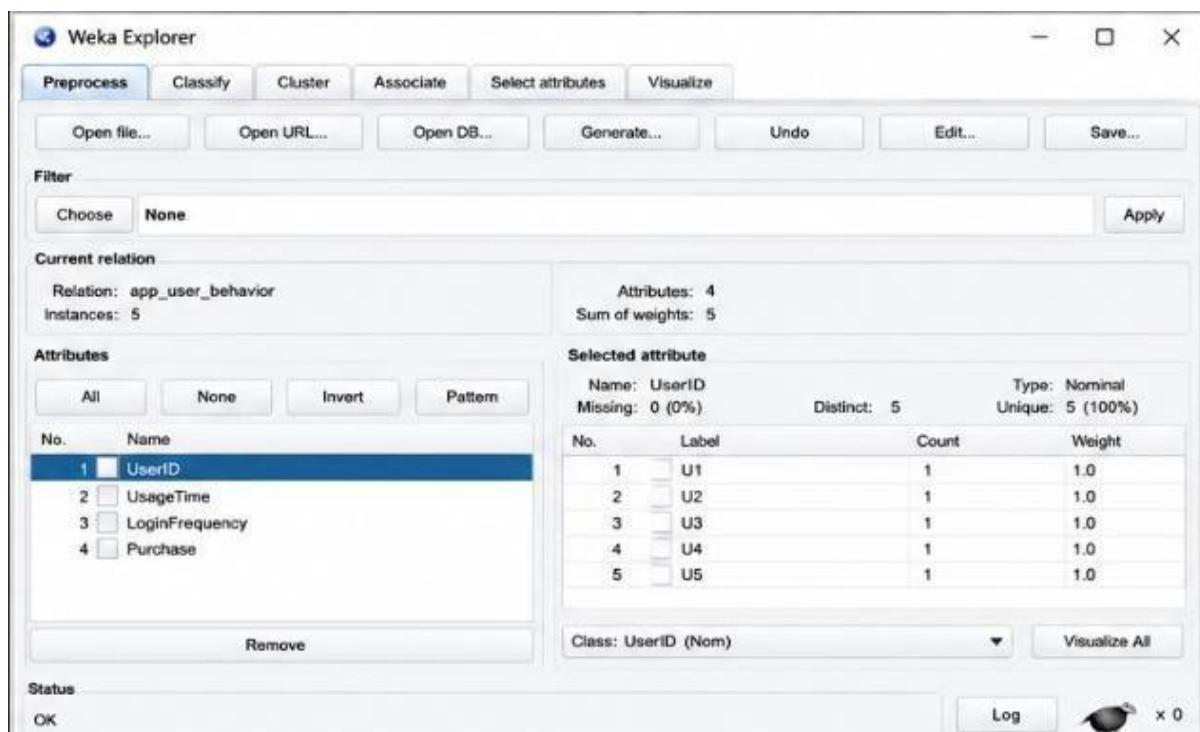
Advantages of Soft Clustering

Soft clustering is useful because:

1. A user can partially belong to multiple behavioral groups.
2. It represents uncertainty more realistically.
3. Users near cluster boundaries can be identified.
4. Marketing strategies can be personalized according to membership probabilities.
5. It helps identify users who may change from one behavior group to another.

Conclusion

Soft clustering provides more flexible customer segmentation than hard clustering because it captures overlapping user behavior and uncertainty.



Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply

Current relation
 Relation: ecommerce_orders
 Instances: 5

Attributes
 All None Invert Pattern

No.	Name
1	ProductCategory
2	OrderValue

Remove

Selected attribute
 Name: ProductCategory
 Missing: 0 (0%) Distinct: 3 Type: Nominal
 Unique: 3 (60%)

No.	Label	Count	Weight
1	Electronics	2	2.0
2	Clothing	2	2.0
3	Furniture	1	1.0

Class: ProductCategory (Nom) Visualize All

Status: OK Log  x 0

Weka Explorer

Preprocess | Classify | **Cluster** | Associate | Select attributes | Visualize

Clusterer
 Choose **SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation
 (Nom) UserID
☒ Store clusters for visualization
 Ignore attributes
 Start Stop

Clusterer output

```

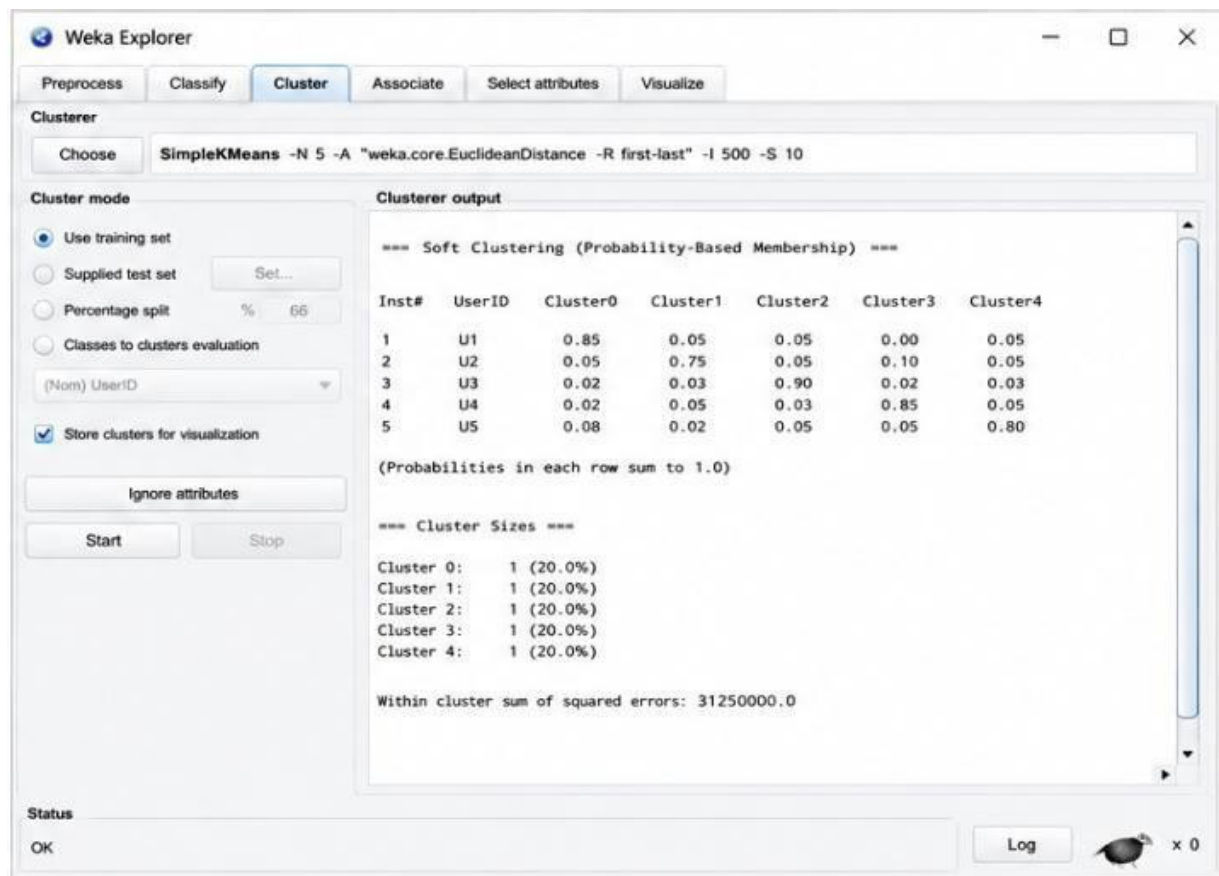
=== Run information ===
Scheme:      weka.clusterers.SimpleKMeans
Relation:    app_user_behavior
Instances:    5
Attributes:   3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster0      Cluster1      Cluster2      Cluster3      Cluster4
                (5.0)          (1.0)         (1.0)         (1.0)         (1.0)
UsageTime      100.0000      120.0000      60.0000      140.0000      50.0000      130.0000
LoginFrequency  13.8000      18.0000      5.0000      22.0000      4.0000      20.0000
Purchase       3200.0000     2000.0000     5000.0000     1500.0000     6000.0000     1800.0000

=== Clustered Instances ===
Inst#  UsageTime  LoginFrequency  Purchase  Cluster
1      120.000    18.000         2000.000    0
2      60.000     5.000          5000.000    1
3      140.000    22.000         1500.000    2
4      50.000     4.000          6000.000    3
5      130.000    20.000         1800.000    4
  
```

Status



Q4. Agricultural Crop Monitoring

Answer

The agricultural dataset contains three attributes:

- Farm rainfall
- Soil fertility score
- Crop yield

The attributes have different scales. Rainfall ranges approximately from **280–520 mm**, soil fertility ranges from **3–9**, and crop yield ranges from **65–90 tons**.

Before Normalization

Without normalization, the rainfall values have a much larger numerical range than soil fertility. Distance-based clustering algorithms may therefore give excessive importance to rainfall.

As a result, farms may be grouped mainly according to their rainfall values rather than considering all three agricultural characteristics equally.

For example:

- F1 and F4 have relatively low rainfall and high soil fertility.
- F2 and F5 have high rainfall and low soil fertility.
- F3 has intermediate values.

After Normalization

Normalization converts the attributes into comparable scales, commonly between **0 and 1**.

After normalization, rainfall, soil fertility, and crop yield contribute more equally to the distance calculation. Therefore, the clusters represent overall agricultural similarity instead of being dominated by rainfall.

Possible groups include:

- **High fertility and high yield farms:** F1 and F4
- **High rainfall and lower fertility/yield farms:** F2 and F5
- **Intermediate agricultural conditions:** F3

Importance of Normalization

Normalization:

1. Prevents large-scale attributes from dominating clustering.
2. Gives each feature a fair contribution.
3. Produces more meaningful clusters.
4. Improves the reliability of distance-based algorithms.
5. Helps researchers make better agricultural decisions.

Conclusion

Normalization is important because it ensures that all agricultural attributes contribute fairly to cluster formation. Therefore, the clusters obtained after normalization are generally more reliable and meaningful.

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... | Open URL... | Open DB... | Generate... | Undo | Edit... | Save...

Filter: Choose **None** Apply

Current relation
Relation: ecommerce_orders
Instances: 5

Attributes
All | None | Invert | Pattern

No.	Name
1	☒ OrderID
2	☐ ProductCategory
3	☐ OrderValue

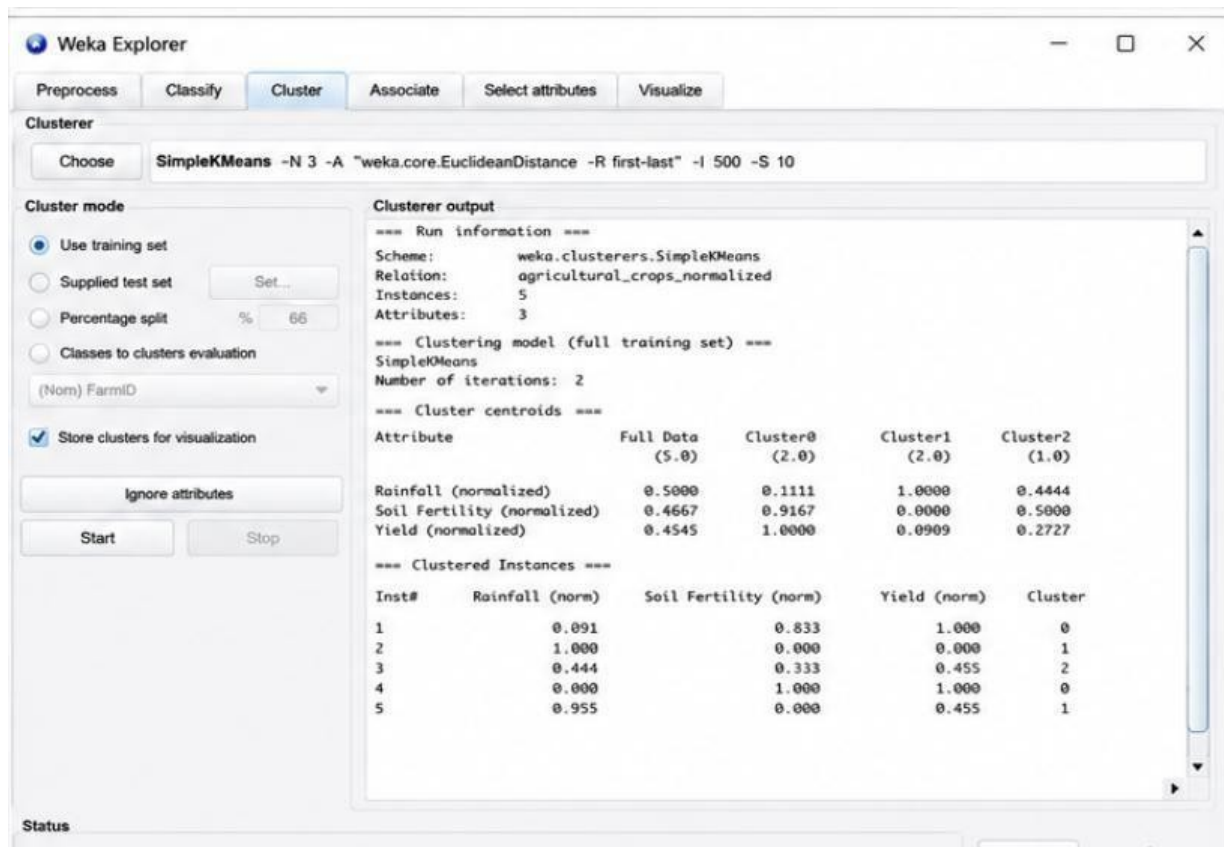
Remove

Selected attribute
Name: OrderID
Missing: 0 (0%)
Distinct: 5
Type: Nominal
Unique: 5 (100%)

No.	Label	Count	Weight
1	O1	1	1.0
2	O2	1	1.0
3	O3	1	1.0
4	O4	1	1.0
5	O5	1	1.0

Class: OrderID (Nom) Visualize All

Status: OK Log x 0



Q5. Employee Performance Evaluation

Answer

The given silhouette scores are:

Number of Clusters Silhouette Score

K = 2	0.48
K = 3	0.68
K = 4	0.52

A silhouette score measures how well data points fit within their own cluster compared with other clusters. A **higher silhouette score indicates better-defined and more separated clusters**.

Among the three models:

- K=2 has a score of **0.48**.
- K=3 has the highest score of **0.68**.
- K=4 has a score of **0.52**.

Therefore, **K = 3 is the most suitable number of clusters**.

Employee Segmentation

The three clusters can represent different performance groups:

- **Cluster 1 – High performers:** High productivity, attendance, and skill scores.
- **Cluster 2 – Average performers:** Moderate productivity, attendance, and skill scores.
- **Cluster 3 – Lower performers:** Lower productivity, attendance, and skill scores.

For example, E1 and E5 have high scores and are likely to belong to the high-performance group. E4 has relatively low scores and may belong to the lower-performance group. E2 and E3 may represent the middle-performance group.

Importance for Employee Evaluation

Using three clusters helps the company:

1. Identify high-performing employees.
2. Identify employees who may need training.
3. Understand different performance patterns.
4. Plan rewards and recognition.
5. Design targeted employee development programs.

Conclusion

The best model is **K = 3** because it has the **highest silhouette score of 0.68**. It provides better-defined employee groups than K=2 or K=4 and can support effective performance evaluation and workforce management.

The screenshot shows the Weka Explorer application with the 'Cluster' tab selected. The 'Clusterer' dropdown is set to 'SimpleKMeans' with parameters: `-N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10`.

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set (Set...)
- ☐ Percentage split (% 66)
- ☐ Classes to clusters evaluation (Nom) Class
- ☒ Store clusters for visualization

Buttons: Ignore attributes, Start, Stop.

Clusterer output:

```
=== Run information ===
Scheme:      weka.clusterers.SimpleKMeans
Relation:    employee_performance
Instances:   5
Attributes:  3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
```

Attribute	Full Data (5.0)	Cluster1 (2.0)	Cluster1 (1.0)
Productivity Score	72.4000	90.0000	72.5000
Attendance (%)	86.4000	95.0000	87.5000
Skill Score	76.4000	86.5000	74.5000

```
=== Clustered Instances ===
```

Inst#	Productivity Score	Attendance (%)	Skill Score	Cluster
1	90.0	95.0	85.0	0
2	75.0	90.0	75.0	1
3	70.0	85.0	74.0	1
4	55.0	70.0	68.0	2
5	95.0	98.0	88.0	0

Status: OK Log x 0

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Clusterer

ChooseSimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

Set...

%66

(Nom) Class

☒ Store clusters for visualization

Ignore attributes

Start

Stop

Clusterer output

=== Run information ===

Scheme: weka.clusterers.SimpleKMeans
Relation: employee_performance
Instances: 5
Attributes: 3

=== Clustering model (full training set) ===

SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===

Attribute	Full Data (5.0)	Cluster0 (2.0)	Cluster1 (2.0)	Cluster2 (1.0)
Productivity Score	72.4000	92.5000	72.5000	55.0000
Attendance (%)	86.4000	96.5000	87.5000	70.0000
Skill Score	76.4000	86.5000	74.5000	68.0000

=== Clustered Instances ===

Inst#	Productivity Score	Attendance (%)	Skill Score	Cluster
1	90.0	95.0	85.0	0
2	75.0	90.0	75.0	1
3	70.0	85.0	74.0	1
4	55.0	70.0	68.0	2
5	95.0	98.0	88.0	0

=== Cluster Evaluation ===

Cluster	Size	SSE	Silhouette Coefficient
0	2	18.5000	0.6823
1	2	22.5000	0.5210
2	1	0.0000	0.0000

Overall Silhouette Coefficient: 0.6142

Status

OK

Log

 x 0